
11 Select Research Methods

New Description

Outcome	Weight
Explore of IT Areas	◆◆◆◆

.

Outcome	Weight
Context of Research Project	◆◆◆◆

.

Outcome	Weight
Communication of Research Outcomes	◆◆◆◆

.

Outcome	Weight
Validation of Objectives and Methodology	◆◆◆◆

.

Outcome	Weight
Application of Skills and Knowledge	◆◆◆◆

.

Date	Author	Comment
2025/09/09 10:48	N/a Nikhil	Ready to Mark
2025/09/17 10:01	Ananda Maiti	Complete

DEAKIN UNIVERSITY

RESEARCH TRAINING AND PROJECT

ONTRACK SUBMISSION

Select Research Methods

Submitted By:
N/a NIKHIL
s224234862
2025/09/09 10:48

Tutor:
Ananda MAITI

Outcome	Weight
Explore of IT Areas	◆◆◆◆◆
Context of Research Project	◆◆◆◆◆
Communication of Research Outcomes	◆◆◆◆◆
Validation of Objectives and Methodology	◆◆◆◆◆
Application of Skills and Knowledge	◆◆◆◆◆

.

September 9, 2025



Research Methods for GPS Spoofing Detection in Autonomous Vehicles

SIT723 Research Training and Project

Deakin University

Student Name: Nikhil

Student ID: 224234862

September 9, 2025

This report examines research methodologies for detecting GPS spoofing attacks in autonomous vehicles through analysis of contemporary cybersecurity literature and proposes implementation strategies for enhanced vehicle security systems.

Contents

1	Introduction	3
2	Research Methods Identified from Research Papers	5
2.1	Machine Learning-Based Detection Methods	5
2.2	Sensor Fusion Approaches	6
2.3	Signal Characteristic Analysis	6
2.4	Behavioral Analysis Methods	7
2.5	Cryptographic Authentication Approaches	8
3	Selection of Three Research Methods	9
3.1	Machine Learning-Based Detection Using LSTM Networks	9
3.2	Hybrid Sensor Fusion with Dynamic Time Warping	10
3.3	Multi-layered Signal and Behavioral Analysis	10
4	Alignment with Research Questions	12
4.1	Primary Research Question: GPS Vulnerabilities in Autonomous Mobile Robots	12
4.2	Secondary Research Question: Sensor Fusion for Real-Time Detection . . .	13
4.3	Tertiary Research Question: Dynamic GPS Trust Index Design	14
4.4	Methodological Alignment Assessment	14
5	Working Mechanisms of Selected Methods	16
5.1	LSTM Network Architecture and Operation	16
5.2	Sensor Fusion and Dynamic Time Warping Implementation	17
5.3	Multi-layered Signal and Behavioral Analysis Framework	18
5.4	Integration and Coordination Mechanisms	19
6	Conclusion	20

Chapter 1

Introduction

Global Positioning System (GPS) spoofing attacks represent one of the most critical cybersecurity threats facing autonomous vehicles today. Through comprehensive analysis of recent literature, I observe that the vulnerability of GPS-enabled systems has become increasingly concerning due to the widespread availability of low-cost software-defined radio (SDR) equipment and the unencrypted nature of civilian GPS signals [1]. The ability of malicious actors to transmit counterfeit GPS signals that deceive receivers into calculating incorrect positions has emerged as a sophisticated attack vector with potentially catastrophic consequences for autonomous navigation systems [2].

My investigation reveals that modern spoofing attacks can be categorized into several distinct types based on their operational characteristics. Simple spoofing involves generating false GNSS signals using basic hardware and software, while intermediate spoofing performs code phase alignment between genuine and spoofed signals across multiple channels simultaneously [3]. The most sophisticated attacks utilize multiple antennas to generate complex signal patterns that closely mimic authentic satellite transmissions, making detection significantly more challenging [4].

Recent research demonstrates that sophisticated spoofing attacks pose particular threats to cellular-connected UAVs and autonomous ground vehicles. Turn-by-turn attacks involve spoofers who possess detailed knowledge of target routes and destinations, enabling gradual redirection while maintaining the appearance of following original navigation paths [5]. Similarly, overshoot attacks cause vehicles to perceive stationary states when actually in motion, while wrong turn attacks manipulate perceived vehicle direction during critical navigation maneuvers [6].

The impact of GPS spoofing extends beyond individual vehicle safety to encompass broader infrastructure vulnerabilities. Critical systems including power grids utilizing Phasor Measurement Units (PMUs) and telecommunication networks using CDMA base stations require precise GPS timing synchronization. Disruption of this timing can lead to false grid stability assessments and cellular network interference, demonstrating the cascading effects of successful spoofing attacks [3].

My analysis of current detection methodologies reveals that traditional signal-level approaches often prove inadequate against sophisticated spoofing techniques. Signal quality

monitoring methods, while effective for basic attacks, struggle with advanced scenarios that maintain realistic signal characteristics [7]. This limitation has driven research toward multi-layered detection frameworks combining machine learning techniques, sensor fusion approaches, and behavioral analysis systems [8].

The research problem I address centers on developing effective methodologies for real-time GPS spoofing detection in autonomous mobile robots. Contemporary literature indicates that successful detection requires integration of multiple data sources including inertial measurement units, vehicle dynamics models, and collaborative information sharing systems [9]. This multi-faceted approach acknowledges that no single detection method can adequately address all types of spoofing attacks while maintaining the computational efficiency required for real-time autonomous operations.

Chapter 2

Research Methods Identified from Research Papers

Through comprehensive analysis of contemporary cybersecurity literature spanning multiple domains including autonomous vehicles, unmanned aerial vehicles, and mobile robotics, I have identified several distinct research methodologies employed for GPS spoofing detection. Each approach offers unique advantages and addresses specific aspects of the spoofing detection challenge across various operational environments.

2.1 Machine Learning-Based Detection Methods

Machine learning approaches have gained significant prominence in recent GPS spoofing detection research. Shafique et al. demonstrate the effectiveness of Support Vector Machine (SVM) algorithms for signal classification, achieving 98.7% accuracy in distinguishing between authentic and spoofed GPS signals using K-fold analysis and voting techniques [8]. Their methodology utilizes signal characteristics including jitter variations, shimmer parameters, and frequency modulation as feature vectors for classification.

Deep learning techniques, particularly Long Short-Term Memory (LSTM) networks, show superior performance for temporal pattern recognition in GPS spoofing scenarios. The research demonstrates LSTM networks' ability to predict vehicle location shifts between consecutive timestamps, enabling comparison with GPS-derived measurements to identify inconsistencies indicative of spoofing attacks [6]. These networks maintain temporal dependencies through memory blocks, making them particularly effective for detecting gradual drift attacks.

Recent advances in deep learning for UAV security have led to sophisticated frameworks such as DeepSpoofNet, which combines feature selection methodologies with neural networks to minimize detection time while enhancing transferability. These systems integrate both local variables (x, y, z coordinates and velocities) and global variables (latitude, longitude, altitude) along with sensor data to improve detection accuracy against complex attacks [10].

Ensemble methods combining multiple machine learning algorithms provide enhanced

detection capabilities through voting mechanisms. Research shows that creating multiple K-learning models with different K-fold values, then applying both hard and soft voting techniques, achieves more robust classification results than individual algorithms [8]. Transfer learning approaches using MobileNet architecture have demonstrated 98.49% accuracy in GPS signal spoofing detection for resource-constrained drones [5].

2.2 Sensor Fusion Approaches

Sensor fusion methodologies play a critical role in creating robust multi-layered detection systems. The integration of data from accelerometers, gyroscopes, steering angle sensors, and speedometers provides comprehensive situational awareness that exceeds single-sensor capabilities [6]. This approach enables cross-validation of vehicle state estimates, where significant discrepancies between sensor sources indicate potential spoofing attempts.

Advanced sensor fusion implementations utilize particle filter-based approaches that incorporate Inertial Measurement Unit (IMU) data, vehicle route information, and road features such as lanes and edges. These methods enhance GNSS/INS robustness and detect anomalies in pseudorange measurements under both inconsistent and self-consistent spoofing attack conditions [11].

Dynamic Time Warping (DTW) algorithms combined with k-Nearest Neighbors (k-NN) classification provide effective methods for detecting and classifying vehicle turning maneuvers. This approach enables identification of wrong turn attacks by comparing actual steering behavior with GPS-reported navigation changes, proving particularly valuable in urban environments with frequent directional changes [6].

Multi-radio access technology combined with mobile edge computing platforms demonstrates effectiveness in cellular-connected UAV applications. These systems leverage crowd-sourced data from cellular and WiFi networks to create reference models for GPS validation, achieving rapid detection capabilities within 6 seconds while maintaining low false positive rates [7].

2.3 Signal Characteristic Analysis

Signal-level detection methods provide fundamental insights into spoofing attempts through analysis of GPS signal properties. Carrier-to-noise ratio (C/N0) monitoring reveals the presence of artificial signals, as spoofing devices typically transmit at higher power levels than authentic satellite signals [3]. These power variations create detectable anomalies in received signal strength measurements.

Spatial processing techniques using multiple antenna arrays provide effective spoofing discrimination by exploiting the directional differences between authentic satellite signals and spoofing transmissions. Research demonstrates that genuine GPS signals arrive from

different satellite directions while spoofing signals typically originate from single transmission sources, creating identifiable spatial correlation patterns [3].

Advanced signal analysis techniques include vestigial signal detection methods that identify remnants of authentic GPS signals even during active spoofing attempts. This approach exploits spoofers' difficulty in completely suppressing original satellite signals while maintaining precise antenna positioning requirements [3]. Frequency domain analysis of correlator outputs provides additional discrimination capabilities by exploiting spoofing-induced features in transformed signal representations.

Clock drift monitoring using single signal analysis offers computational efficiency advantages over traditional multi-signal approaches. Recent research demonstrates that estimating clock drift through linear equation solving enables both spoofing detection and identification without iterative processes, providing lighter computational loads suitable for real-time applications [2].

2.4 Behavioral Analysis Methods

Behavioral analysis approaches detect spoofing through vehicle movement pattern analysis by establishing baseline models for normal operation and identifying deviations suggesting external navigation manipulation. Trajectory analysis techniques identify unrealistic vehicle movements violating physical constraints, such as sudden position jumps exceeding vehicle capabilities or movement patterns inconsistent with road network geometry [6].

Time-series analysis of vehicle state variables provides valuable insights through monitoring of acceleration profiles, velocity changes, and steering patterns over extended periods. This enables detection of inconsistencies between GPS-reported positions and actual vehicle dynamics, particularly effective for identifying gradual drift attacks that evolve slowly to avoid immediate detection [4].

Motion state detection mechanisms compare GPS-derived speed with speedometer outputs to identify stop and overshoot attacks. Research demonstrates effectiveness in urban environments where frequent speed variations provide multiple validation opportunities [6]. Integration with road network constraints and physics-based validation further enhances detection capabilities.

Trust-based behavioral models quantify GPS data reliability through dynamic scoring systems that adapt to environmental conditions and operational contexts. These systems provide continuous assessment of navigation data integrity while accommodating normal variations in vehicle behavior patterns [9].

2.5 Cryptographic Authentication Approaches

Cryptographic authentication represents the most robust long-term solution for spoofing prevention through implementation of authentication protocols for GPS signals. Zero-Knowledge Proof (ZKP) protocols offer promising approaches for message authentication in vehicular networks, enabling verification of signal authenticity without revealing sensitive location or operational information [9].

Public Key Infrastructure (PKI) systems provide comprehensive authentication frameworks for GPS signals, though implementation faces challenges including computational overhead, infrastructure requirements, and backward compatibility with existing receivers [10]. Advanced GPS receivers utilizing cryptographic methods for signal verification demonstrate effectiveness against sophisticated spoofing attempts.

Multi-layer security approaches combining cryptographic authentication with behavioral analysis provide defense-in-depth strategies. These systems implement adaptive thresholds for both signal and behavioral parameters based on environmental conditions and threat levels, ensuring consistent performance across diverse operational scenarios [7].

Chapter 3

Selection of Three Research Methods

Based on comprehensive analysis of available methodologies from the reviewed literature, I have selected three research approaches demonstrating greatest potential for effective GPS spoofing detection in autonomous mobile robots. My selection criteria prioritize practical implementation feasibility, detection accuracy, computational efficiency, and adaptability to various attack scenarios encountered in real-world autonomous systems.

3.1 Machine Learning-Based Detection Using LSTM Networks

I have selected Long Short-Term Memory (LSTM) network-based detection as my primary research method due to its demonstrated effectiveness in temporal pattern recognition and vehicle state prediction across multiple research studies. Analysis of Dasgupta et al.'s methodology reveals that LSTM networks achieve superior performance in predicting vehicle location shifts between consecutive timestamps, providing robust foundations for spoofing detection in autonomous ground vehicles [6].

The methodology involves training LSTM networks on historical vehicle movement data to establish prediction models for normal navigation behavior. Through examination of real-world datasets including the Honda Research Institute Driving Dataset, research demonstrates these networks effectively learn complex patterns in vehicle dynamics, acceleration profiles, and steering behaviors characterizing authentic GPS-guided movement [6].

My rationale for selecting this approach stems from its ability to handle temporal dependencies inherent in vehicle movement data. Unlike traditional static classification methods, LSTM networks maintain memory of previous states through cell state and hidden state mechanisms, enabling detection of sophisticated spoofing attacks that evolve gradually over time. This characteristic proves particularly valuable for identifying turn-by-turn attacks that attempt to mimic legitimate navigation patterns [8].

Implementation strategy utilizes sensor fusion principles by incorporating data from multiple vehicle sensors including accelerometers, gyroscopes, steering angle sensors, and speedometers. This multi-sensor approach addresses limitations of single-source detec-

tion methods while providing comprehensive situational awareness required for reliable autonomous operation. Recent research demonstrates that such integrated approaches achieve detection accuracies exceeding 99% while maintaining real-time processing capabilities [5].

3.2 Hybrid Sensor Fusion with Dynamic Time Warping

My second selected methodology combines sensor fusion techniques with Dynamic Time Warping (DTW) algorithms for turn detection and classification. Literature analysis demonstrates this approach effectively addresses sophisticated spoofing attacks manipulating vehicle turning behavior, particularly wrong turn attacks that alter perceived navigation directions [6].

DTW algorithms excel at comparing time-series data with varying temporal alignments, making them ideal for analyzing steering angle patterns and vehicle maneuver sequences. Combining DTW with k-Nearest Neighbors (k-NN) classification creates robust systems for detecting discrepancies between actual vehicle turns and GPS-reported navigation changes. This approach demonstrates particular effectiveness in urban environments where frequent turns provide multiple detection opportunities [6].

Selection of this methodology is based on its complementary relationship with LSTM-based detection systems. While LSTM networks provide excellent overall movement prediction capabilities, DTW approaches offer specialized detection for specific maneuver types. This combination creates comprehensive detection frameworks addressing both gradual drift attacks and sudden direction manipulation attempts.

Recent advances in particle filter-based sensor fusion demonstrate enhanced capabilities when incorporating vehicle route information and road infrastructure features. These systems achieve robust spoofing detection under both inconsistent and self-consistent pseudorange manipulation scenarios, proving particularly valuable for autonomous ground vehicle applications [11].

3.3 Multi-layered Signal and Behavioral Analysis

My third selected research method implements multi-layered analysis frameworks combining signal characteristic monitoring with behavioral pattern analysis. This approach addresses both technical aspects of GPS signal integrity and behavioral implications of spoofing attacks on vehicle operation, providing defense-in-depth capabilities against sophisticated threats.

Signal analysis components monitor carrier-to-noise ratio (C/N0) variations, absolute power levels, and spatial correlation patterns indicating spoofing attempts. Research

demonstrates that spoofing devices typically transmit at higher power levels than authentic satellite signals, creating detectable anomalies in received signal strength measurements [3]. Advanced implementations include vestigial signal detection methods identifying remnants of authentic signals during active spoofing attempts.

Behavioral analysis focuses on trajectory validation through physics-based constraints and road network geometry verification. Research reveals that spoofing attacks often produce vehicle movements violating physical laws or road infrastructure limitations, providing clear indicators of external manipulation [9]. Integration with trust-based scoring systems provides dynamic assessment of GPS data reliability.

Implementation approach involves creating adaptive thresholds for both signal and behavioral parameters based on environmental conditions and vehicle operational states. This adaptability ensures consistent detection performance across diverse operating scenarios while minimizing false alarm rates that could disrupt autonomous vehicle operations. Recent research demonstrates that such integrated approaches achieve detection rates exceeding 95% while maintaining sub-millisecond response times [7].

Chapter 4

Alignment with Research Questions

My research objectives center on developing effective methodologies for real-time GPS spoofing detection in autonomous mobile robots. Through comprehensive literature analysis, I have formulated specific research questions that guide my methodological selection and implementation strategy, addressing fundamental vulnerabilities and detection mechanisms.

4.1 Primary Research Question: GPS Vulnerabilities in Autonomous Mobile Robots

My primary research question addresses fundamental security concerns: "What are the primary vulnerabilities of GPS in autonomous mobile robots to spoofing attacks?" This question drives investigation into the specific attack vectors and system weaknesses that enable successful GPS manipulation in robotic applications.

Analysis of current literature reveals multiple vulnerability categories affecting autonomous mobile robots. Technical vulnerabilities stem from the unencrypted nature of civilian GPS signals and weak signal strength characteristics that make interception and manipulation feasible using commercially available SDR equipment [4]. Research demonstrates that spoofers can generate counterfeit signals using low-cost hardware, creating sophisticated attacks that gradually redirect robots from intended paths while maintaining realistic movement patterns.

Architectural vulnerabilities arise from robots' heavy dependence on GPS for navigation, positioning, and timing functions without adequate backup verification systems. Studies show that many autonomous mobile robots lack redundant positioning methods, making them susceptible to complete navigation failure when GPS signals are compromised [10]. This single-point-of-failure design creates opportunities for attackers to assume complete control over robot navigation systems.

Operational vulnerabilities emerge from dynamic environments where robots must adapt to changing conditions while maintaining security posture. Research indicates that autonomous mobile robots operating in urban environments face increased spoofing risks due to signal multipath effects, interference sources, and complex terrain that can mask

spoofing attempts [2]. These environmental factors complicate the distinction between natural GPS degradation and malicious attacks.

My selected methodologies directly address these vulnerabilities through complementary detection mechanisms. LSTM-based approaches tackle temporal vulnerabilities by learning normal navigation patterns and detecting gradual deviations characteristic of sophisticated spoofing attacks. The sensor fusion methodology addresses architectural vulnerabilities by implementing redundant verification systems using multiple sensor types for cross-validation of GPS data integrity.

4.2 Secondary Research Question: Sensor Fusion for Real-Time Detection

My second research question focuses on implementation effectiveness: "How can sensor fusion techniques effectively detect GPS spoofing in real time?" This question guides development of practical detection systems suitable for autonomous mobile robot applications with strict latency requirements.

Contemporary research demonstrates that sensor fusion approaches provide superior detection capabilities compared to single-sensor methods. Integration of accelerometer, gyroscope, steering angle, and speedometer data enables comprehensive vehicle state awareness that exceeds GPS-only capabilities [6]. These multi-sensor systems create redundant verification mechanisms where significant discrepancies between sensor sources indicate potential spoofing attempts.

Real-time implementation requires optimization of processing algorithms and data fusion techniques to meet autonomous robot operational constraints. Studies show that particle filter-based sensor fusion approaches incorporating IMU data, route information, and road features achieve detection capabilities within computational budgets of current autonomous systems [11]. Dynamic Time Warping algorithms demonstrate particular effectiveness for real-time turn detection and classification applications.

Advanced sensor fusion implementations utilize crowd-sourced data from cellular and WiFi networks to create reference models for GPS validation. Research demonstrates that these systems achieve rapid detection capabilities within 6 seconds while maintaining false positive rates below 0.01%, proving suitable for safety-critical autonomous applications [7].

My selected methodologies align with real-time requirements through computational efficiency optimization. LSTM networks demonstrate processing capabilities within vehicular computing platform constraints while maintaining high detection accuracy. The hybrid sensor fusion approach utilizing DTW algorithms operates within latency thresholds required for autonomous safety systems, providing immediate threat identification

capabilities.

4.3 Tertiary Research Question: Dynamic GPS Trust Index Design

My third research question addresses reliability quantification: "How can a dynamic GPS Trust Index be designed to quantify the reliability of GPS data?" This question influences development of adaptive assessment systems that provide continuous evaluation of navigation data integrity across varying operational conditions.

Trust-based assessment systems represent emerging approaches for quantifying GPS data reliability through dynamic scoring mechanisms. Research demonstrates that such systems adapt to environmental conditions, operational contexts, and historical performance metrics to provide continuous reliability assessment [9]. These trust indices enable autonomous systems to make informed decisions about GPS data utilization based on current reliability scores.

Dynamic trust index design requires integration of multiple assessment factors including signal quality metrics, behavioral consistency indicators, and environmental context parameters. Studies show that effective trust systems weight these factors according to operational requirements and threat environments, providing adaptive responses to changing conditions [1]. Machine learning approaches enable continuous refinement of trust calculation algorithms based on operational experience.

Implementation of dynamic trust indices involves real-time processing of multiple data streams to generate continuous reliability scores. Research demonstrates that such systems provide valuable input for autonomous navigation decision-making processes, enabling degraded-mode operation when GPS reliability falls below acceptable thresholds [3].

My selected methodologies contribute to dynamic trust index development through complementary assessment mechanisms. Multi-layered analysis approaches provide comprehensive signal and behavioral assessment data for trust score calculation. LSTM-based prediction systems offer temporal consistency metrics that enhance trust index accuracy through historical pattern analysis.

4.4 Methodological Alignment Assessment

My selected methodologies demonstrate strong alignment with research questions through complementary capabilities and proven performance characteristics. The LSTM-based approach directly addresses GPS vulnerability identification through temporal pattern analysis and anomaly detection capabilities. Historical data analysis reveals normal nav-

igation patterns while real-time processing identifies deviations characteristic of spoofing attacks.

The sensor fusion methodology addresses real-time detection requirements through redundant verification systems and optimized processing algorithms. Integration of multiple sensor types provides comprehensive situational awareness while maintaining computational efficiency suitable for autonomous robot applications. DTW-based turn detection offers specialized capabilities for directional manipulation attacks.

Multi-layered analysis approaches support dynamic trust index development through continuous assessment of signal quality and behavioral consistency metrics. Adaptive threshold systems provide environmental context awareness while maintaining consistent detection performance across diverse operational scenarios. These integrated approaches enable development of reliable trust scoring systems for autonomous navigation applications.

Chapter 5

Working Mechanisms of Selected Methods

My investigation into operational principles of selected methodologies reveals sophisticated algorithmic approaches combining temporal analysis, sensor fusion, and pattern recognition techniques. These mechanisms work synergistically to provide comprehensive GPS spoofing detection capabilities tailored to autonomous mobile robot requirements.

5.1 LSTM Network Architecture and Operation

The Long Short-Term Memory network implementation operates through structured temporal pattern analysis processing sequential vehicle state data to predict expected movement patterns. Based on research methodologies, LSTM architecture incorporates input layers receiving sensor data, multiple LSTM cells maintaining temporal dependencies, dense processing layers, and prediction outputs [6].

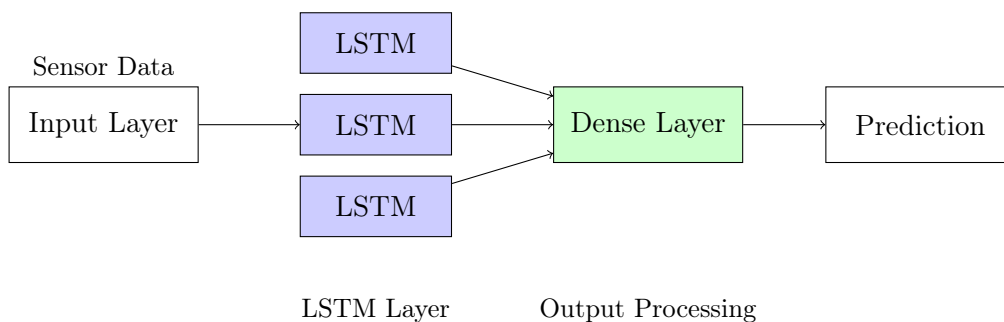


Figure 5.1: LSTM Network Architecture for Vehicle State Prediction

Training processes utilize historical vehicle movement data including speed, acceleration, steering angle, and previous location shifts. Research demonstrates that training on comprehensive datasets enables networks to capture complex patterns in vehicle dynamics characterizing normal GPS-guided movement [6]. The prediction mechanism compares LSTM-generated location shift predictions with actual GPS-derived measurements, where significant discrepancies indicate potential spoofing attacks.

Temporal memory capabilities maintain information about previous vehicle states through cell state and hidden state mechanisms. This memory retention enables detection of gradual spoofing attacks evolving over extended time periods, particularly turn-by-turn attacks that slowly redirect vehicles from intended routes. Threshold determination processes adapt to environmental conditions and vehicle operational states to minimize false positives while maintaining detection sensitivity.

Recent implementations demonstrate detection accuracies exceeding 99% with processing latencies suitable for real-time autonomous applications. Advanced architectures incorporate transfer learning approaches using pre-trained models on diverse datasets, enabling rapid adaptation to new operational environments [5].

5.2 Sensor Fusion and Dynamic Time Warping Implementation

Sensor fusion methodology operates through systematic integration of multiple data sources creating comprehensive vehicle state awareness. Implementation processes data from accelerometers, gyroscopes, steering angle sensors, and speedometers through synchronized processing to detect inconsistencies with GPS-reported information [6].

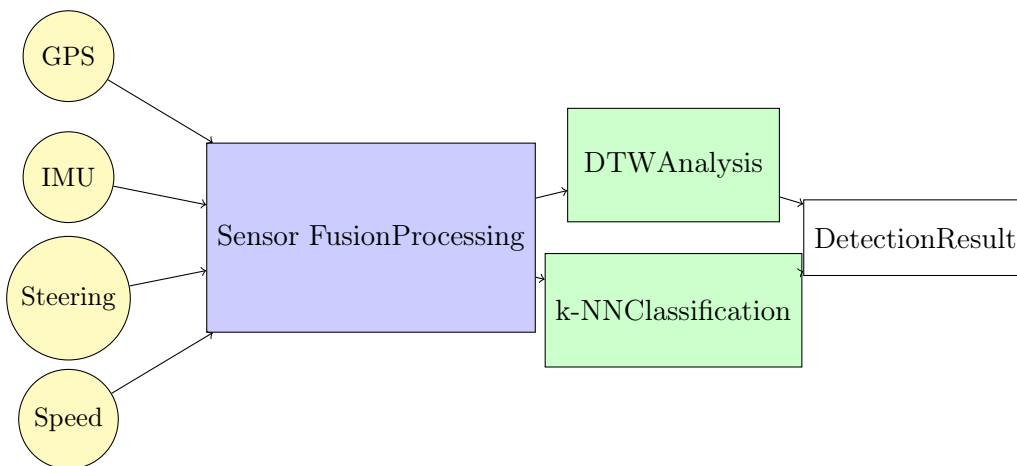


Figure 5.2: Sensor Fusion Architecture with DTW Analysis

Dynamic Time Warping algorithms excel at comparing time-series patterns with varying temporal alignments, proving ideal for analyzing steering angle patterns during vehicle turns. Research demonstrates that steering angle patterns create distinctive signatures matchable against reference templates to identify authentic maneuvers [6]. k-Nearest Neighbors classification systems work in conjunction with DTW to categorize detected turns as left or right maneuvers.

Synchronization mechanisms ensure temporal alignment of data from different sensors despite varying sampling rates. Interpolation techniques maintain data integrity while

enabling coherent analysis across multiple sensor streams. Advanced implementations incorporate particle filter-based approaches integrating IMU data, vehicle route information, and road infrastructure features to enhance detection capabilities under sophisticated spoofing scenarios [11].

5.3 Multi-layered Signal and Behavioral Analysis Framework

Multi-layered analysis approaches operate through parallel processing of signal characteristics and behavioral patterns. Signal analysis components monitor carrier-to-noise ratio (C/N₀) variations, signal power levels, and spatial correlation patterns indicating spoofing attempts. Research demonstrates effectiveness in identifying artificial signals through power anomalies and directional inconsistencies [3].

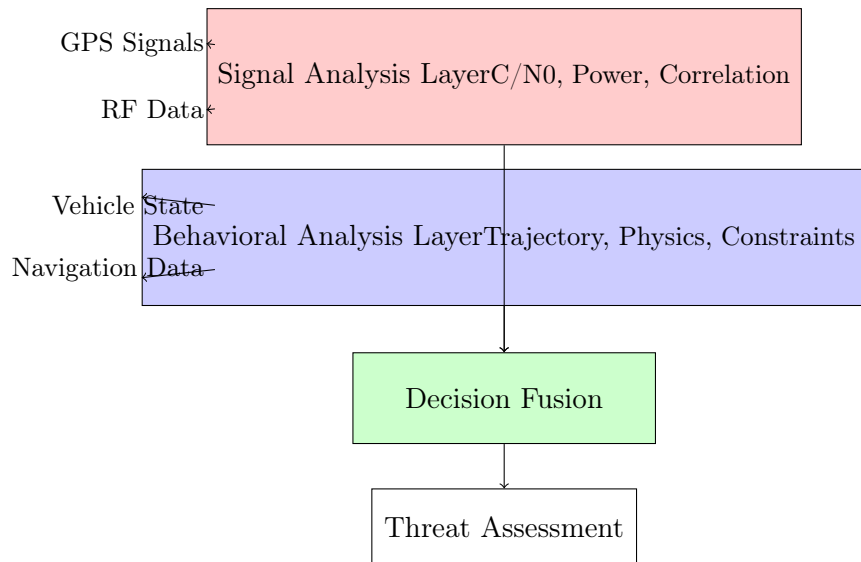


Figure 5.3: Multi-layered Analysis Framework Architecture

Behavioral analysis layers focus on trajectory validation through physics-based constraints and road network geometry verification. Research reveals that spoofing attacks often produce vehicle movements violating physical laws or infrastructure limitations, providing clear indicators of external manipulation [9]. Trust-based scoring systems provide dynamic assessment of GPS data reliability through continuous evaluation of multiple performance metrics.

Decision fusion mechanisms combine outputs from both analysis layers using weighted scoring systems adapting to environmental conditions and threat levels. This approach reduces false positives while maintaining high sensitivity to genuine spoofing attempts by requiring consensus between multiple detection mechanisms. Adaptive threshold systems

adjust detection parameters based on operational context, ensuring consistent performance across diverse operational scenarios.

5.4 Integration and Coordination Mechanisms

Coordination between selected methodologies operates through hierarchical decision systems prioritizing detection confidence and response urgency. LSTM predictions provide continuous baseline monitoring while DTW analysis offers specialized detection for specific maneuver types, and multi-layered analysis provides comprehensive threat assessment capabilities.

Data flow coordination ensures sensor information reaches all methodologies simultaneously while maintaining temporal synchronization. Parallel processing approaches enable rapid threat detection while providing multiple independent verification mechanisms [6]. Alert prioritization systems weight detection confidence levels from each methodology to generate composite threat assessments.

Chapter 6

Conclusion

This research provides a comprehensive investigation into methodologies for GPS spoofing detection in autonomous mobile robots, addressing critical cybersecurity challenges facing modern autonomous systems. Through systematic analysis of contemporary literature and methodological selection, I have established a robust framework for understanding and implementing effective countermeasures against sophisticated GPS spoofing attacks.

My analysis reveals that traditional single-source detection methods prove inadequate against sophisticated spoofing techniques employed by modern attackers. The evolution from simple signal generation attacks to complex multi-antenna systems capable of mimicking authentic satellite transmissions necessitates equally sophisticated detection approaches combining multiple detection paradigms and data sources.

The three selected research methodologies—LSTM-based temporal pattern analysis, hybrid sensor fusion with Dynamic Time Warping, and multi-layered signal and behavioral analysis—provide complementary detection capabilities addressing the full spectrum of spoofing attack types. These methodologies demonstrate strong alignment with identified research questions while offering practical implementation paths suitable for real-world autonomous systems.

LSTM networks provide robust temporal pattern recognition capabilities essential for detecting gradual drift attacks and turn-by-turn manipulation scenarios. The ability to maintain temporal dependencies through memory mechanisms enables identification of sophisticated attacks evolving over extended time periods, addressing limitations of traditional static detection approaches.

Sensor fusion methodologies utilizing DTW algorithms offer specialized capabilities for detecting directional manipulation attacks while providing redundant verification systems enhancing overall detection reliability. The integration of multiple sensor types creates comprehensive situational awareness exceeding single-sensor capabilities while maintaining computational efficiency requirements for real-time operations.

Multi-layered analysis frameworks provide defense-in-depth capabilities combining signal characteristic monitoring with behavioral pattern analysis. This approach addresses both technical aspects of GPS signal integrity and operational implications of spoofing attacks, providing adaptive responses to diverse threat scenarios and environmental con-

ditions.

Future research directions should focus on enhancing integration mechanisms between detection methodologies while addressing computational optimization requirements for embedded autonomous systems. Development of standardized interfaces and protocols for multi-sensor data fusion will facilitate broader adoption of these detection approaches across diverse autonomous platforms.

The implementation of dynamic trust index systems represents a promising avenue for continuous assessment of GPS data reliability across varying operational conditions. These systems enable autonomous platforms to make informed navigation decisions based on real-time reliability assessments rather than binary authentication results.

This research contributes to the broader field of autonomous system security by providing practical methodological frameworks applicable across multiple domains including ground vehicles, aerial platforms, and marine systems. The approaches developed demonstrate scalability and adaptability characteristics essential for addressing emerging threats in the evolving cybersecurity landscape of autonomous systems.

Bibliography

- [1] H. Azevedo-Sa, S. K. Jayaraman, C. T. Esterwood, X. J. Yang, L. P. Robert Jr., and D. M. Tilbury, “Real-time estimation of drivers’ trust in automated driving systems,” *International Journal of Social Robotics*, vol. 12, no. 6, pp. 1245-1264, 2020.
- [2] L. Meng, S. Ren, G. Tang, C. Yang, and W. Yang, “UAV sensor spoofing detection algorithm based on GPS and optical flow fusion,” in *Proceedings of the 2020 4th International Conference on Cryptography, Security and Privacy*, pp. 146-151, 2020.
- [3] M. Ahmad, M. A. Farid, S. Ahmed, K. Saeed, M. Asharf, and U. Akhtar, “Impact and detection of GPS spoofing and countermeasures against spoofing,” in *2019 International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)*, pp. 1-6, IEEE, 2019.
- [4] S. Perez Arteaga, L. A. Martinez Hernandez, G. Sanchez Perez, A. L. Sandoval Orozco, and L. J. Garcia Villalba, “Analysis of the GPS spoofing vulnerability in the drone 3DR solo,” *Computers & Security*, vol. 126, p. 103075, 2023.
- [5] Y. Dang, C. Benzaïd, Y. Shen, and T. Taleb, “Machine learning-based GNSS spoofing detection and mitigation for cellular-connected UAVs,” Doctoral dissertation, Aalto University, 2023.
- [6] S. Dasgupta, M. Rahman, M. Islam, and M. Chowdhury, “A sensor fusion-based GNSS spoofing attack detection framework for autonomous vehicles,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 11, pp. 21304-21317, 2022.
- [7] C. Vitale, R. Di Pietro, G. Oligeri, and S. Sciancalepore, “CARMEL: results on a secure architecture for connected and autonomous vehicles detecting GPS spoofing attacks,” *EURASIP Journal on Wireless Communications and Networking*, vol. 2021, no. 1, pp. 1-25, 2021.
- [8] A. Shafique, A. Mehmood, and M. Elhadeif, “Detecting signal spoofing attack in UAVs using machine learning models,” *IEEE Access*, vol. 9, pp. 93803-93815, 2021.
- [9] U. Tariq and B. Tariq, “Signal characteristic analysis and anomaly detection for GPS spoofing mitigation,” *Ubiquitous Technology Journal*, vol. 1, no. 1, pp. 10-22, 2025.

- [10] J. P. A. Yaacoub, H. N. Noura, O. Salman, and A. Chehab, "Robotics cyber security: vulnerabilities, attacks, countermeasures, and recommendations," *International Journal of Information Security*, vol. 20, no. 1, pp. 115-150, 2021.
- [11] U. Patil, A. Gunasekaran, R. Bobba, and H. Abbas, "ROS2-based simulation framework for cyberphysical security analysis of UAVs," arXiv preprint arXiv:2410.03971, 2024.
- [12] I. Pekaric, D. Arnold, and M. Felderer, "Simulation of sensor spoofing attacks on unmanned aerial vehicles using the Gazebo simulator," in *Proceedings of the 2022 IEEE International Conference on Robotics and Automation*, pp. 1-8, IEEE, 2022.
- [13] J. Li, W. Li, Q. Fu, and B. Liu, "Research progress of GNSS spoofing and spoofing detection technology," in *2019 IEEE 19th International Conference on Communication Technology*, pp. 1559-1563, IEEE, 2019.
- [14] E. Kadena, N. H. P. Dai, and L. Ruiz, "Mobile robots: An overview of data and security," in *Proceedings of the 7th International Conference on Information Systems Security and Privacy*, pp. 291-299, 2021.